SIMULATING EMERGENT FLOCKING BEHAVIOR

This presentation will explore behavior of a group of agents (e.g., birds or fish) based on three simple rules:
Separation, **Alignment**, and **Cohesion**.

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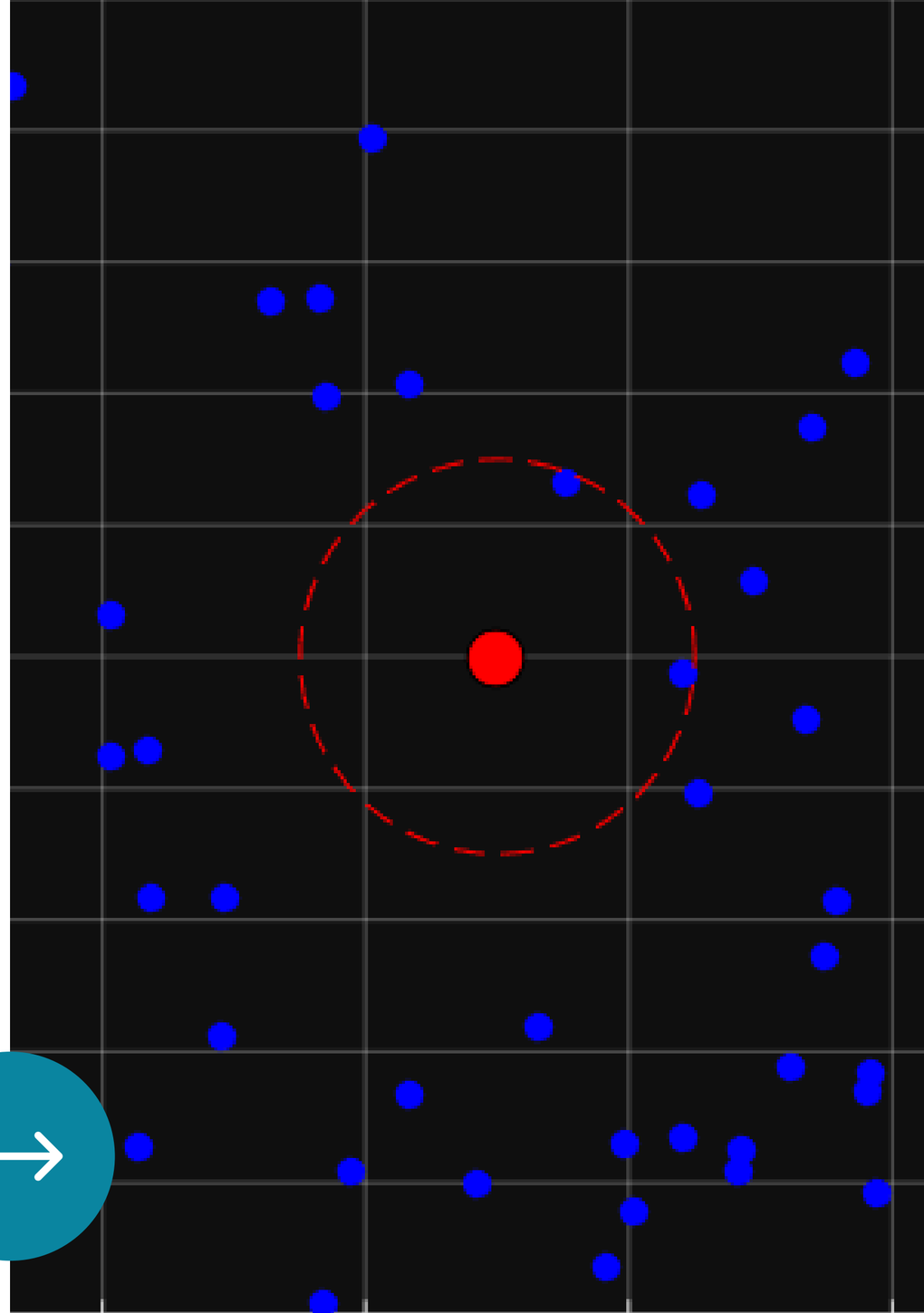


PROBLEM DESCRIPTION

Objective: To simulate the movement of autonomous agents (birds/fish) without using a central controller or leader.

Core Concept: Based on Craig Reynolds' 1986 "Boids" model, demonstrating Emergent Behavior—complex group dynamics arising from simple local rules.

The Three Governing Rules:

01

Separation

Avoid collisions with close neighbors (Personal Space).

02

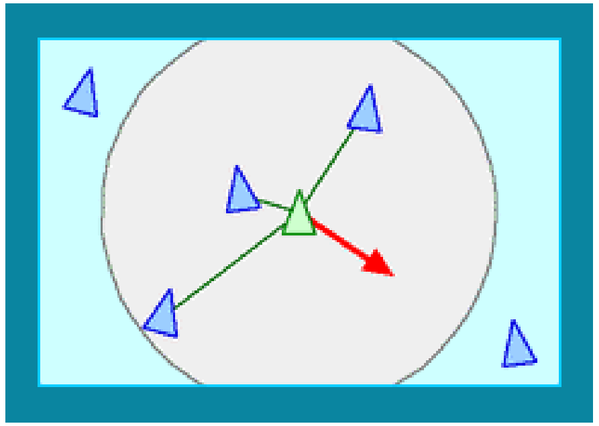
Alignment

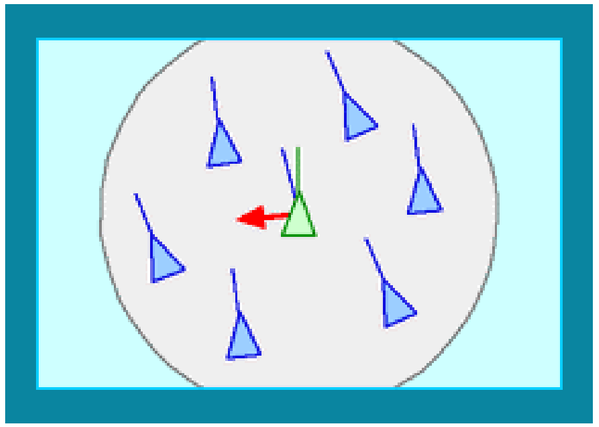
Steer in the same direction as neighbors (Synchronous movement).

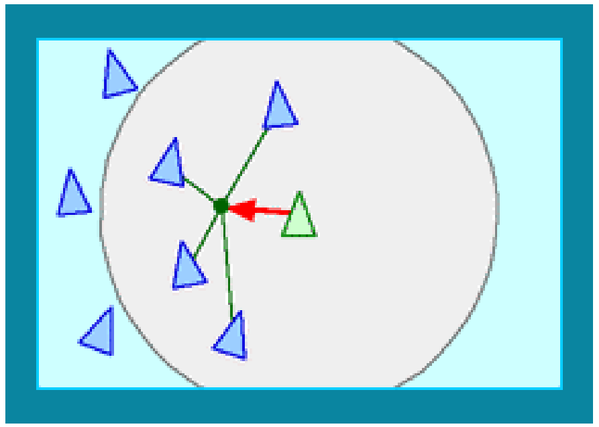
03

Cohesion

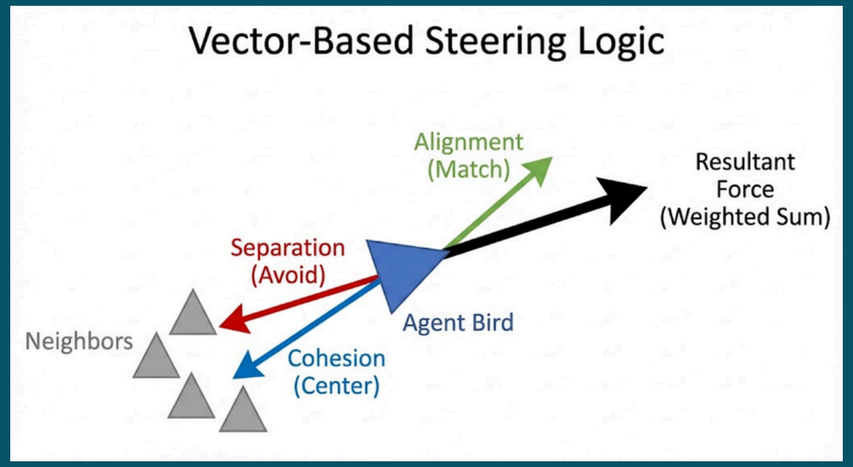
Steer toward the center of the local group (Staying together).







THE "BRAIN" OF THE BOID



01

Local Sensing (Input):

- Each agent is autonomous.
- It only detects neighbors within a defined VisualRange (blind to the rest of the world).

02

Vector Calculation (Process):

- We calculate three directional vectors for every neighbor:
- **Separation**: Vector pointing away to avoid crowding.
- **Alignment**: Vector matching the neighbor's velocity.
- **Cohesion**: Vector pointing to the neighbor's position.

03

Weighted Summation (Decision):

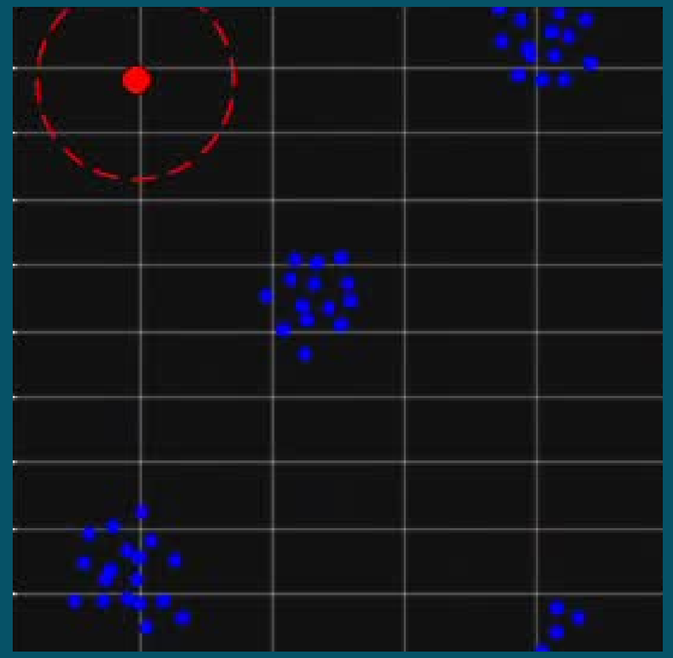
- The final direction is a weighted sum of all three forces:

$$\vec{F}_{total} = (w_1 \times Sep) + (w_2 \times Align) + (w_3 \times Coh)$$

04

Physics Update (Action):

- The result is added to the current velocity and capped at MaxSpeed.



ANALYSIS

A

Emergence Verified:

- From random initialization (t=0), organized flocking stabilizes within 50 time steps.
- Proves that global complexity arises from local rules (no leader needed).

B

Parameter Sensitivity (What happens if we break the math?):

Parameter (w)	Condition	Observed Behavior	Analogy
Separation (w_1)	High (↑)	Flocking breaks down; agents scatter and refuse to group.	High-Pressure Gas
Cohesion (w_3)	High (↑)	Agents collapse into a single, unnatural dense point.	Black Hole Effect
Alignment (w_2)	High (↑)	Movement becomes perfectly parallel and rigid; no organic turning.	Military March

C

Predator Dynamics:

- Flash Expansion: The flock successfully executes an escape maneuver, splitting rapidly to creating a "safety void" around the predator.
- Re-cohesion: Once the threat passes, the cohesion rule naturally pulls the separated groups back together.

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Final Status Report:
Boids: 50 (Alive and Flocking)
Predator: 1 (Still Hungry)
Simulation: Completed!
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